SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	23/10/2026	**Duration:**	60 minutes

Code of Conduct

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

1. Study and Analyze the Scenario

1.1 Business Domain

The project belongs to the Environmental Management and E-Waste Recycling domain. It aims to manage electronic waste collection, transportation, classification, recycling, and sustainability reporting through a centralized software platform.

1.2 Stakeholders

- Consumers: Submit e-waste collection requests and track pickups.
- Collection Centres: Receive, classify, weigh, and store e-waste.
- Logistics Providers: Schedule pickups and transport materials.
- Recyclers: Process e-waste and record recovered materials.
- Administrators: Manage users, requests, and system operations.
- Government Authorities: Monitor compliance and environmental reports.
- Development and Operations Teams: Develop, test, deploy, and maintain the system.

1.3 Existing Challenges

- Manual collection requests and record maintenance.
- Poor coordination between collection centres and recyclers.
- Difficulty tracking e-waste during transportation.
- Inaccurate or incomplete recycling records.
- Lack of centralized sustainability reports.
- Limited visibility into collection and recovery performance.

1.4 System Objectives

1. Automate e-waste collection and pickup scheduling.
2. Track materials from collection to recycling.
3. Maintain classification and recycling records.
4. Generate sustainability dashboards and reports.
5. Improve security, scalability, and system reliability.
6. Reduce manual work and encourage responsible disposal.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

ID	Functional Requirement
FR1	The system shall allow users to register, log in, and log out.
FR2	The system shall allow consumers to submit e-waste collection requests.

FR3	The system shall allow consumers to select available pickup slots.
FR4	The system shall allow administrators to assign collection requests to logistics providers.
FR5	The system shall allow logistics providers to update pickup and delivery status.
FR6	The system shall allow collection centres to classify and record e-waste weight.
FR7	The system shall allow recyclers to record recycling and material recovery activities.
FR8	The system shall maintain a history of e-waste movement.
FR9	The system shall generate sustainability reports and dashboards.
FR10	The system shall allow administrators to manage users and system records.
FR11	The system shall provide notifications for important request and pickup updates.
FR12	The system shall maintain application logs and monitoring information.
FR13	The system shall support secure APIs and automated software testing and deployment.

2.2 Non-Functional Requirements

Requirement	Description
Performance	Normal user requests should respond within 3 seconds under the agreed test load.
Security	Use secure authentication, role-based access, encryption, and input validation.
Reliability	Prevent loss of confirmed requests and maintain consistent records.
Usability	Provide a simple, responsive interface for all stakeholder groups.
Scalability	Support increasing users, requests, and transaction volumes.
Maintainability	Use modular code, documentation, and automated tests.
Availability	Target 99.5% monthly availability, excluding planned maintenance.
Data Integrity	Validate required fields and maintain consistent material records.
Portability	Support deployment through containers in compatible environments.
Monitoring	Record errors, service health, and performance metrics.

Note: Performance and availability values are proposed targets and should be confirmed with stakeholders before implementation.

3. Identify Actors and Use Cases

3.1 Primary and Secondary Actors

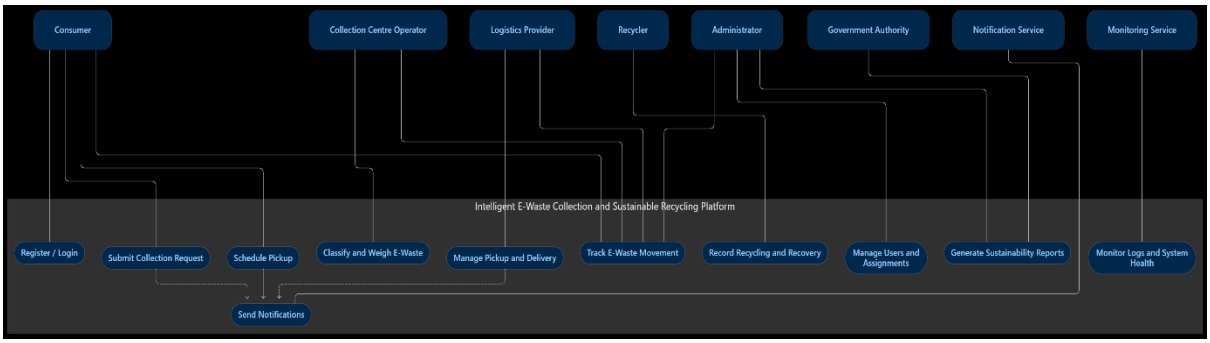
Actor	Type	Main Use Cases
Consumer	Primary	Register, log in, submit collection request, schedule

		pickup, track status.
Collection Centre Operator	Primary	Receive e-waste, classify materials, record weight, update storage details.
Logistics Provider	Primary	View assigned pickups, update collection and delivery status.
Recycler	Primary	Record recycling processes and recovered materials.
Administrator	Primary	Manage users, assign requests, monitor operations, generate reports.
Government Authority	Secondary	Review authorized compliance and sustainability reports.
Notification Service	Secondary	Send pickup confirmations and status notifications.
Monitoring Service	Secondary	Collect application logs, health checks, and performance metrics.

3.2 Major Use Cases

- Consumer: Register/Login, Submit Request, Schedule Pickup, Track Request.
- Collection Centre: Receive E-Waste, Classify Materials, Record Weight.
- Logistics Provider: View Assignments, Collect E-Waste, Update Delivery Status.
- Recycler: Record Recycling, Update Recovered Materials.
- Administrator: Manage Users, Assign Pickups, Monitor Operations, Generate Reports.
- Government Authority: View Compliance and Sustainability Reports.
- External Services: Send Notifications and Monitor System Health.

3.3 Use Case Diagram



Note: This is a high-level use case representation. The dashed arrows indicate that relevant workflows may trigger notifications.

4. Software Requirement Specification (SRS)

4.1 Introduction

Purpose: The purpose of this SRS is to define the functional and non-functional requirements of the Intelligent E-Waste Collection and Sustainable Recycling Platform.

Intended audience: Project developers, testers, administrators, project managers, collection-centre operators, logistics providers, and recyclers.

Product perspective: The platform is a centralized web-based application supported by a

database, secure APIs, containerized services, monitoring, and reporting tools.

4.2 Problem Description

Increasing electronic waste creates challenges in collection, transportation, material classification, and recycling. Existing processes may rely on manual records and disconnected systems, resulting in poor coordination and limited traceability.

The proposed platform integrates stakeholders and records each stage of the e-waste lifecycle, from collection requests to recycling and material recovery.

4.3 Scope of the System

Included:

- User accounts and role-based access.
- Collection requests and pickup scheduling.
- E-waste classification and weight recording.
- Transportation and status tracking.
- Recycling and material recovery records.
- Sustainability reports and dashboards.
- Notifications, secure APIs, logging, and monitoring.

Excluded from the initial version:

- Physical collection and transportation operations.
- Direct control of recycling machinery.
- Guaranteed automatic recognition of every e-waste item.
- Automatic legal certification without authorized review.
- Advanced GPS tracking if compatible tracking devices and integrations are unavailable.

4.4 Use Case Descriptions

Use Case 1: Submit E-Waste Collection Request

Field	Description
Use Case ID	UC-01
Actor	Consumer
Preconditions	Consumer is logged in.
Main Flow	Enter device type, quantity, address, contact details; submit request; system validates and stores details; system generates a request ID.
Alternative Flow	If details are missing or invalid, the system displays an error message.
Postconditions	A collection request is stored with an initial status.

Use Case 2: Schedule Pickup

Field	Description
Use Case ID	UC-02
Actor	Consumer
Preconditions	A valid collection request exists.
Main Flow	View available slots, select a date and time, confirm the booking, and receive confirmation.
Alternative Flow	If the slot is unavailable, the system asks the consumer to select another slot.
Postconditions	The pickup schedule is linked to the collection request.

Use Case 3: Track E-Waste Movement

Field	Description

Use Case ID	UC-03
Actor	Consumer, Collection Centre Operator, Administrator
Preconditions	A valid request or tracking ID exists.
Main Flow	Open the tracking page, retrieve the latest status, and display recorded movement updates.
Alternative Flow	If the ID is invalid, display a suitable error message.
Postconditions	The user views the latest available tracking information.

Use Case 4: Classify E-Waste

Field	**Description**
Use Case ID	UC-04
Actor	Collection Centre Operator
Preconditions	The operator is authenticated and the received material is registered.
Main Flow	Select the material category, enter weight and condition, validate details, and save the record.
Alternative Flow	Reject missing or invalid values and request correction.
Postconditions	Classification and weight records are stored.

Use Case 5: Record Recycling Activity

Field	**Description**
Use Case ID	UC-05
Actor	Recycler
Preconditions	The recycler is authorized and the received batch is recorded.
Main Flow	Enter batch details, processed quantity, recovered materials, and processing status; save the record.
Alternative Flow	Reject invalid quantities or inconsistent batch information.
Postconditions	The recycling record is saved and linked to the corresponding batch.

Use Case 6: Generate Sustainability Report

Field	**Description**
Use Case ID	UC-06
Actor	Administrator, authorized Government Authority
Preconditions	The actor has report access and relevant data exists.
Main Flow	Select a reporting period, retrieve collection and recycling data, calculate indicators, and display the report.
Alternative Flow	Display a no-data message when no records match the selected period.
Postconditions	The report is displayed and can be exported if export functionality is enabled.

5. Data Requirements

The system shall maintain the following database entities.

Entity	Important Attributes
User	User ID, name, email, password hash, role, contact details
Collection Request	Request ID, user ID, device type, quantity, address, status
Pickup Schedule	Schedule ID, request ID, date, time, assigned provider
E-Waste Batch	Batch ID, category, weight, condition, current location
Transport Record	Transport ID, batch ID, provider ID, pickup time, delivery status
Recycling Record	Record ID, batch ID, processed quantity, recovered materials
Sustainability Report	Report ID, reporting period, collected weight, recycled weight
Notification	Notification ID, recipient, message, timestamp, delivery status
Audit Log	Log ID, user ID, activity, timestamp, result

Data Validation Rules

- Each user, request, batch, and recycling record must have a unique identifier.
- Email addresses must follow a valid format.
- Weight and quantity must be positive values.
- A pickup must reference an existing collection request.
- Recycling records must reference a valid batch.
- Only authorized users can modify protected records.
- Collection, transport, and recycling updates must maintain a traceable history.

6. External Interface Requirements

6.1 User Interface

- Responsive web interface for desktop, tablet, and mobile.
- Login and registration pages.
- Collection request and pickup scheduling forms.
- Tracking dashboard.
- Collection-centre and recycler dashboards.
- Administrative and sustainability reporting dashboards.

6.2 Software Interfaces

- Frontend: HTML, CSS, JavaScript.
- Backend: PHP or another suitable server-side framework.
- Database: MySQL.
- API: RESTful APIs using HTTPS and JSON.
- Deployment: Docker containers and a compatible hosting environment.
- CI/CD: Automated build, testing, and deployment pipeline.
- Monitoring: Application logs, error reports, and health checks.

These are proposed technologies and can be adjusted to match the project's implementation requirements.

6.3 Communication Interfaces

- HTTPS for secure client-server communication.
- Email or SMS gateway for pickup notifications, if configured.
- GPS or mapping API for location tracking, if integrated.
- Secure API authentication for external service communication.

7. System Features

Feature	Description	Expected Result
User Management	Register and authenticate users.	Secure platform access.
Collection Management	Submit and manage collection requests.	Organized collection process.
Pickup Scheduling	Select slots and assign logistics providers.	Improved coordination.
E-Waste Tracking	Record and display movement status.	Better traceability.
Classification	Record categories, quantities, and weights.	Accurate material information.
Recycling Management	Record processing and recovered materials.	Improved recycling records.
Sustainability Analytics	Calculate collection and recycling indicators.	Environmental performance visibility.
Notifications	Inform users about important updates.	Better communication.
System Monitoring	Record errors and monitor service health.	Improved reliability.

8. Assumptions and Constraints

8.1 Assumptions

1. Users have access to a device and internet connection.
2. Stakeholders provide accurate collection and recycling information.
3. Collection centres and recyclers are registered before using operational features.
4. Administrators verify user roles and permissions.
5. Logistics providers update transportation status regularly.
6. External notification and mapping services are available when integrated.
7. Users follow applicable e-waste handling and disposal regulations.

8.2 Constraints

Constraint Type	Description
Technical	Limited infrastructure, third-party API availability, and compatibility requirements.
Operational	Collection schedules depend on staff, vehicles, and centre capacity.
Economic	Budget limits hosting, development, maintenance, and integration options.
Legal	Data protection, environmental regulations, and applicable e-waste rules must be followed.
Time	Development must be completed within the assigned

	academic project period.
Data	Inaccurate or missing stakeholder updates can reduce reporting accuracy.
Security	Sensitive user information must be protected from unauthorized access.

9. Validate Requirement Completeness and Consistency

Requirement validation ensures that the SRS is clear, complete, consistent, and testable.

9.1 Validation Checklist

Validation Area	Verification
Completeness	All major stakeholder roles and core workflows are included.
Consistency	Collection, transportation, and recycling statuses follow a defined workflow.
Clarity	Requirements use specific terms such as “shall” and avoid vague statements.
Testability	Requirements have measurable acceptance criteria or verification methods.
Non-redundancy	Duplicate requirements are combined or referenced consistently.
Feasibility	Proposed features match the available budget, technology, and project duration.
Security	Access permissions and protection of sensitive data are defined.
Traceability	Each requirement has an ID and can be linked to use cases and tests.

9.2 Potential Gaps and Improvements

- Status consistency: Define standard statuses such as Requested, Scheduled, Collected, Received, Processing, Recycled, and Cancelled.
- Data accuracy: Validate weights, quantities, and batch identifiers at entry.
- Access control: Define permissions separately for consumers, operators, logistics providers, recyclers, and administrators.
- Error recovery: Specify backup, restoration, and recovery procedures.
- Notifications: Define what triggers a notification and how failed deliveries are handled.
- Performance: Confirm the expected number of users, concurrent requests, and response-time targets.
- Reporting accuracy: Define formulas for recycled weight, recovery rate, and reporting periods.
- Privacy: Define data retention, deletion, and access policies according to applicable laws.
- Testing: Map every functional requirement to at least one test case.
- Traceability: Maintain a requirement-to-use-case-to-test mapping for future changes.

9.3 Requirement Traceability Example

Requirement ID	Use Case	Verification Method
FR1	Registration and Login	Authentication tests
FR2	Submit Collection Request	Form and API tests
FR3	Schedule Pickup	Slot validation tests
FR5	Update Delivery Status	Integration tests

FR6	Classify E-Waste	Data validation tests
FR7	Record Recycling	Database and workflow tests
FR9	Generate Sustainability Reports	Report calculation tests
FR12	Monitor Application	Log and health-check tests

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5

Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25